

RESEARCH ARTICLE

Swift heavy ion-induced defect formation and electronic structure evolution in Zr-doped CeO₂ (Ce_{0.8}Zr_{0.2}O₂)

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Abstract

This study investigates the effects of 100 MeV iodine ion irradiation on zirconia-doped ceria (Ce_{0.8}Zr_{0.2}O₂) to simulate fission fragment damage in nuclear fuel at fluences up to 5×10^{13} ions/cm². X-ray photoelectron spectroscopy confirms the partial reduction of Ce⁴⁺ to Ce³⁺ and Zr⁴⁺ to Zr³⁺/Zr²⁺, driven by irradiation-induced charge trapping and stabilized by oxygen vacancies. These localized electrons distort the lattice, forming small polarons that govern the redox equilibrium and contribute to defect tolerance. UV–Vis diffuse reflectance spectroscopy confirms narrowing of the band gap (3.42–2.85 eV) and increase in Urbach energy from 0.93 to 1.85 eV confirmed enhancement of electronic disorder and band tailing. Photoluminescence measurements reveal a progressive quenching of emission intensity with increasing fluence, associated with nonradiative recombination via polaron-stabilized defect states. The combined spectroscopic results suggest that both electronic excitation and nuclear energy loss induce a synergistic mechanism involving polaron formation and oxygen Frenkel pair creation. This dual pathway enables defect accommodation without structural collapse. These results demonstrate that Ce_{0.8}Zr_{0.2}O₂ resists amorphization through a balance of irradiation-induced redox activity and native nonstoichiometry, making it a strong candidate for radiation-resistant ceramics in nuclear and aerospace applications, and a mechanistic model for defect-tolerant oxides.

KEYWORDS

photoluminescence, SHI irradiation, UV–Vis diffuse reflectance spectroscopy, X-ray photoelectron spectroscopy

1 | INTRODUCTION

Ceramic oxides are crucial for transmuting and immobilizing minor actinides produced in nuclear reactors. Their physicochemical properties are significantly affected under intense radiative environments due to radiation-induced defect dynamics. Swift heavy ion (SHI) irradiation, particularly in the energy range of hundreds of MeV generates significantly damaged amorphous regions,

defect clusters or point defects including oxygen Frenkel pairs, depending on the oxide structure. These defects play a vital role in controlling material behavior under radiation.^{1–3} SHI irradiation is a potential technique for tailoring the structural, electrical, and optical properties of materials including metals, semiconductors, insulators, and polymers by rapidly depositing high electronic energy over ultrashort timescales. This energy excites the lattice and induces significant alterations in material

properties.⁴ Understanding the interplay between defect formation and mechanical response is essential for developing radiation-resistant materials for advanced nuclear applications.⁵

Uranium dioxide (UO_2) is the most common material employed as nuclear fuel. Numerous theoretical and experimental investigations have been conducted on it encompassing various aspects. Several dopants including titanium, niobium, magnesium, chromium, and zirconium oxides, have been extensively explored to improve the thermophysical, mechanical, and chemical properties of the fuel matrix, as well as to reduce fission gas release and swelling during irradiation.⁶ ZrO_2 -doped UO_2 ceramic pellets have been considered as high-performance nuclear fuel in light water reactors (LWRs) because of their enhanced characteristics.⁷ These cover better corrosion resistance in high-temperature water or steam, steady thermal conductivity during irradiation, and a reduced swelling rate relative to pure uranium dioxide fuels during in-pile irradiation.^{6,8,9} Zirconia (ZrO_2) and ceria (CeO_2) having cubic phase shares structural similarities with nuclear fuel materials like (UO_2 , PuO_2 , and ThO_2) making them promising candidates for inert matrix materials (IMMs). They demonstrate elevated melting points, less neutron absorption, higher thermal conductivity, enhanced thermal stability, insolubility in coolant, compatibility with cladding materials, ease of manufacturing processes, and radiation stability.^{10,11}

Recent studies have irradiated CeO_2 with ions of varying masses and energies to simulate radiation damage analogous to that experienced by UO_2 fuels due to fission fragments.^{3,12,13} Zhang et al.¹⁴ performed a critical analysis of recent literature on Zr-doped ceria for catalytic applications and employed density functional theory (DFT) for explaining the impact of Zr^{4+} doping. Therefore, fluorite-structured solid solutions of ZrO_2 - CeO_2 oxides are of particular importance as model phases for studying the stability and properties of analogous systems like ZrO_2 - UO_2 and ZrO_2 - PuO_2 dioxide, theoretically explain by Weck et al.¹⁵ and experimentally explained by Singh et al.² Our prior research thoroughly examined the radiation sensitivity and structural evolution of complex oxides under severe circumstances. In $\text{Gd}_2\text{Ti}_2 - y\text{Zr}_y\text{O}_7$ pyrochlores, we exhibited thermally activated damage recovery and compositional influences on phase transitions, emphasizing improved radiation endurance with elevated zirconium content.^{16–20} Nonstoichiometric $\text{Nd}_{1.8}\text{Zr}_{2.2}\text{O}_{7.1}$ and nanostructured $\text{Nd}_2\text{Zr}_2\text{O}_7$ demonstrated enhanced resistance to electronic excitation attributed to defect recovery mechanisms and nanoscale characteristics.^{16,18} Short-range ordering during irradiation in $\text{Gd}_2\text{Ti}_2 - y\text{Zr}_y\text{O}_7$ demonstrated that compositional adjustment is essential for structural durability.¹⁷

X-ray photoelectron spectroscopy (XPS) can throw light on the composition of cerium ions (Ce^{3+} and Ce^{4+}) present in CeO_2 . SHI irradiation on CeO_2 / CeO_2 -based mixed oxides can cause partial reduction of Ce^{4+} to Ce^{3+} and generates oxygen vacancies.^{12,21} Consequently, an advanced technique is needed to quantitatively evaluate the microchemistry of CeO_2 -based oxides, which directly impacts aqueous durability.²² XPS experiences difficulties in detecting cerium oxidation states in materials due to the complicated characteristics of valence and core electron spectra.²³ The Ce 3d XPS spectra of Ce^{3+} ions can reveal satellites with intensities higher than the one of the primary peaks.²⁴ The spectra of CeO_2 are particularly complex, with most literature suggesting a mixed valence state containing both Ce^{3+} and Ce^{4+} ions,²⁵ emphasizing the need for careful interpretation of the spectral data.

Research on SHI irradiation effects on CeO_2 - ZrO_2 systems is limited, but investigations of similar materials like Y_2O_3 - ZrO_2 (YSZ) provide valuable insights.²⁶ YSZ does not amorphize even at high radiation doses, instead undergoing stress, lattice swelling, and mechanical property enhancements due to ion irradiation.^{3,13,27} Similar experiments with CeO_2 have demonstrated expansion in lattice parameters and alterations in oxidation states without resulting in bulk amorphization or phase transformation, Ohno et al.²⁷ reported the reduction of Ce^{4+} to Ce^{3+} in CeO_2 under 200 MeV Xe ions irradiation, while Ishikawa et al.²⁸ observed lattice contraction in CeO_2 upon irradiation with 150 MeV and 200 MeV ions. Tracy et al. found linear unit cell expansion in bulk CeO_2 subjected to irradiation using 167 MeV ^{132}Xe and 950 MeV ^{197}Au ions.³ Cureton et al. noticed unit cell expansion, microstrain development, and alterations in oxidation states in CeO_2 and UO_2 without bulk amorphization or phase transformation, after irradiation with 946 MeV Au^+ ions.¹³ These findings suggest that while CeO_2 does not undergo amorphization or structural transformation, it does experience a reduction into trivalent ions, similar to UO_2 , upon energetic ion irradiation. This resilience highlights the potential of the CeO_2 - ZrO_2 system as a model for inert matrix fuels, where ZrO_2 serves as the IMM and CeO_2 acts as a surrogate for Pu.¹²

This study examines the impact of fission-energy ion radiation on the nonstoichiometry and structural integrity of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, which is isostructural with UO_2 and PuO_2 , and is also a potential candidate for the development of inert fuel matrices capable of containing high-activity nuclear waste. XPS along with photoluminescence (PL) investigations display significant chemical and electronic modifications under irradiation, even though the material resists amorphization and retains its fluorite structure up to high ion fluences. Importantly, unlike single-cation CeO_2 or ZrO_2 , $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ exhibits

coupled redox behavior of Ce/Zr cations under SHI irradiation which provides advanced defect tolerance through dynamic equilibrium between native nonstoichiometry and irradiation defects, linked to oxygen vacancy formation and polaronic charge localization. These synergistic effects enhance defect accommodation and radiation tolerance, underscoring the relevance of this system for nuclear applications.

2 | EXPERIMENTAL DETAILS

2.1 | Synthesis and swift heavy ion irradiation

Polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ powder (space group $Fm\bar{3}m$) was synthesized at 1500°C using high-purity CeO_2 ($<50\text{ nm}$, $>99.95\%$) and ZrO_2 ($<5\text{ }\mu\text{m}$, $>99.00\%$) powders from Sigma Aldrich, forming $8 \times 1\text{ mm}^2$ pellets. Sample preparation details are provided in our previous work.²⁹

Irradiation experiments were carried out at the Inter-University Accelerator Centre (IUAC), New Delhi, using 100 MeV I^{7+} ions at fluences ranging from 1×10^{12} to $5 \times 10^{13}\text{ ions/cm}^2$. A uniform irradiation over a $10 \times 10\text{ mm}^2$ area was achieved using a magnetic scanner. The experiments were performed at room temperature under high vacuum ($\sim 1 \times 10^{-7}\text{ mbar}$) to suppress surface oxidation and contamination. SRIM-2013 simulations by Singh et al.² confirm that the projected ion range ($\sim 7.9\text{ }\mu\text{m}$) lies well within the pellet thickness, with energy deposition dominated by electronic excitations. Displacement energy values of 25 eV for Ce, 25 eV for Zr, and 28 eV for O were used, consistent with literature on fluorite oxides under ion irradiation.²⁹

2.2 | Characterization

XPS was performed using a PHI 5000 Versa Probe III (Ulvac-Phi, Inc.) under an ultrahigh vacuum of 10^{-7} mbar with a monochromatic Al K_α X-ray source (1.48 keV , $100\text{ }\mu\text{m}$ beam diameter). XPS probes to a depth of approximately $5\text{--}10\text{ nm}$, which is sufficient to capture chemical changes resulting from irradiation. Sputtering was deliberately avoided to prevent surface modification and preserve the original chemical states. Surface reoxidation is minimal due to low oxygen mobility at room temperature. Spectral acquisition utilized precise scan parameters (step width, pass energy), a stage tilt angle of 45° , and a neutralizer (1.5 V , $20\text{ }\mu\text{A}$) to compensate for surface charging. The recorded spectra were deconvoluted using XPSPeak Fit 4.1 with Shirley background subtraction, and peak fitting was performed using a Gaussian–Lorentzian

sum function to accurately resolve overlapping peaks and chemical states. UV–Vis diffuse reflectance spectroscopy (UVDRS) was conducted on an Agilent Cary 5000 spectrophotometer ($175\text{--}2500\text{ nm}$) equipped with a double out-of-plane Littrow monochromator, tungsten–halogen and deuterium light sources, an R928 photomultiplier for UV–Vis, and a Peltier-cooled PbS detector for near-infrared (NIR), enabling temperature-dependent reflectance and absorbance measurements for powders and thin films. The spectra were processed using the instrument's integrated software, which directly converts reflectance to absorbance, which is based on the Kubelka–Munk function (K–M). PL spectra were recorded using a Jasco FP 8300 spectrofluorometer with a modified Rowland mount (1650 lines/mm concave grating monochromators). A 150 W xenon flash lamp provided broadband excitation ($200\text{--}750\text{ nm}$), while a high-sensitivity photomultiplier ensured accurate emission detection. Emission spectra ($200\text{--}750\text{ nm}$) were averaged over ≥ 5 scans to reduce intensity fluctuations and enhance the signal-to-noise ratio for high-precision spectral analysis.

3 | RESULTS AND DISCUSSION

3.1 | Spectroscopic analysis of pristine $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$

3.1.1 | X-ray photoelectron spectroscopy analysis

XPS was utilized to examine the chemical states of Ce, Zr, and O in polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ before and after irradiation as shown in Figure 1, providing insights into local coordination and oxidation states through analysis of Ce 3d, Zr 3d, and O 1s core levels. All XPS spectra were calibrated to the C 1s peak at 284.8 eV for accurate comparison.

Figure 1A displays the O 1s spectrum of the pristine sample, where the peak at $\sim 529.02\text{ eV}$ corresponds to lattice oxygen in the fluorite $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ structure, associated with Ce^{4+} ions.³⁰ The $\sim 531.3\text{ eV}$ peak is linked to hydroxyl,³¹ highly polarized oxygen near defect sites,⁴⁷ or oxygen vacancies in metal oxides.^{12,33} Under the given experimental conditions, the absence of hydroxyl and carbonate species suggests that the $\sim 531.3\text{ eV}$ feature originates from oxygen vacancies. The $\sim 530\text{ eV}$ peak corresponds to Ce^{3+} in Ce_2O_3 , indicating partial cerium reduction.³⁴ Thus, even in the pristine sample, the presence of Ce^{3+} along with oxygen vacancies is evident.

Figure 1B shows the Ce 3d peaks split into Ce $3d_{5/2}$ (v_0, v_1, v_2, v_3, v_4) and Ce $3d_{3/2}$ (u_0, u_1, u_2, u_3, u_4) ionizations. The v_0, v_2, u_0 , and u_2 peaks correspond to Ce^{3+} , while

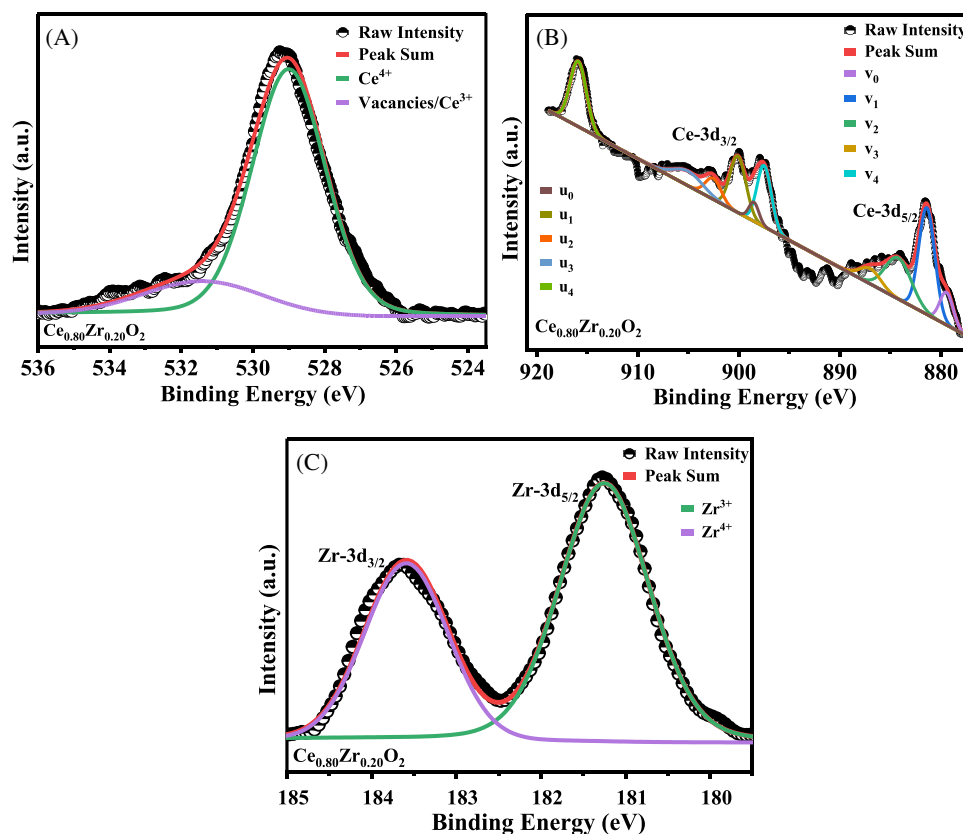


FIGURE 1 X-ray photoelectron spectroscopy (XPS) spectra of pristine polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ pellets, including (A) O 1s spectra, providing insights into oxygen states and bonding; (B) Ce 3d spectra, highlighting cerium's electronic and oxidation states, crucial for the $\text{Ce}^{4+}/\text{Ce}^{3+}$ ratio; and (C) Zr 3d spectra, revealing zirconium's chemical states and interactions within the ceria matrix.

v_1 , v_3 , v_4 , u_1 , u_3 , and u_4 indicate Ce^{4+} states.³⁵ This indicates that Ce^{4+} is the predominant oxidation state on the surface, with a minor presence of Ce^{3+} . The Ce^{3+} peaks suggest slight oxygen deficiency in the samples. The Ce^{3+} state arises from the spontaneous reduction of Ce^{4+} (0.97 Å) to Ce^{3+} (1.10 Å), mitigating lattice contraction induced by the smaller Zr^{4+} ions (0.84 Å).³⁶

The Zr 3d region (Figure 1C) shows peaks at 181.4 eV and 183.8 eV, attributed to the Zr 3d_{5/2} and Zr 3d_{3/2}, spin-orbit components of Zr^{4+} , respectively.^{11,37,38} No evidence of lower valence zirconium species is observed, indicating the predominance of Zr^{4+} in the unirradiated sample.³⁹

3.1.2 | UV-Vis diffuse reflectance spectroscopy analysis

Diffuse reflectance spectra of pristine $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ were recorded in the 200–800 nm range and converted into absorption spectra using the K–M function (Figure 2), α or $F(R) = \frac{(1-2R)^2}{2R}$, where R is the measured reflectance. The K–M transformation provides a pseudoabsorbance proportional to the absorption coefficient and is widely applied

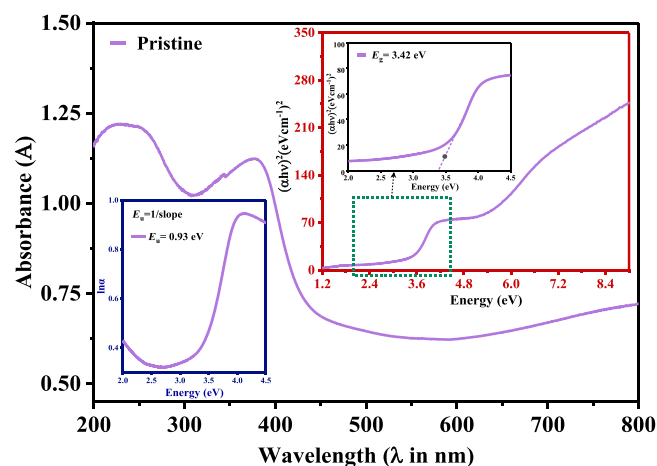


FIGURE 2 The absorption spectra of pristine polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, whereas $(\alpha h\nu)^2$ versus $h\nu$ Tauc's plot in red inset and $\ln(\alpha)$ versus $h\nu$ for Urbach energy in the blue inset.

to oxide ceramics for extracting optical band gaps. The absorbance spectra of the synthesized $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ solid solution exhibit two distinct absorption bands at ~245 nm (5.06 eV) and ~375 nm (3.42 eV), consistent with previous literature.^{40–43}

These bands are attributed to electronic transitions from the O 2p orbital of the valence band (VB) to the Ce 5d orbital of Ce⁴⁺ in the conduction band (CB), and from the O 2p orbital of the VB to the localized Ce 4f orbital of Ce⁴⁺ in the CB, as also observed in other studies.^{40,44} The UV absorption in CeO₂ arises from charge-transfer transitions between the O 2p and Ce 4f states in O²⁻ and Ce⁴⁺, which is much stronger than the 4f¹–5d¹ transition in Ce³⁺ species within the mixed-valence CeO₂ system.^{41,42} The UVDRS spectra, recorded from 200–800 nm, confirm that material exhibits a direct band gap with sharp absorption at the band edge.

The energy band gap (E_g) was calculated using Tauc's equation, derived from the Schuster–Kubelka–Munk (K–M) function approximation,^{40,44,45}

$$(\alpha h\nu)^{\frac{1}{m}} = \beta (h\nu - E_g) \quad (1)$$

In this equation, α is the linear absorption coefficient, h is Planck's constant, ν is the photon frequency, E_g represents the band gap energy, and β is a constant related to the band properties. The exponent m , which reflects the transition mode, varies with the material's crystallinity.

The band gap of Ce_{0.8}Zr_{0.2}O₂ was determined by plotting the Schuster–Kubelka–Munk function, $(\alpha h\nu)^{1/m}$ against photon energy ($h\nu$) for both indirect ($m = 2$) and direct ($m = 0.5$) transitions. For allowed transitions, extrapolating the $(\alpha h\nu)^2$ versus $h\nu$ plot to zero ordinate provides an estimate of the direct optical band gap. The estimated value of the band gap is 3.42 eV in good agreement, which falls within the range of the reported values for CeO₂ ($E_g \sim 3.15$ eV) by Nadja et al.,⁴⁰ $E_g \sim 3.28$ eV for Ce_{0.8}Zr_{0.2}O₂ by Bhabu et al.,⁴² $E_g \sim 3.45$ eV for CeO₂ Leel et al.,⁴⁶ $E_g \sim 3.42$ eV for CeO₂,⁴³ and $E_g \sim 4.49$ eV for CeO₂ by Babitha et al.⁴⁴ The E_g values are also calculated by the inverse logarithmic derivative method and presented in Table 4.⁴⁶ The term $d[\ln(\alpha h\nu)]/d[h\nu]$ can be plotted versus $h\nu$ and the maximum of the curve gives the value of band gap. Therefore, Equation (1) can be rewritten as:

$$\ln(\alpha h\nu)^{\frac{1}{m}} = \ln(h\nu - E_g) \quad (2)$$

$$\ln(\alpha h\nu) = m \ln(h\nu - E_g) \quad (3)$$

$$\frac{d[\ln(\alpha h\nu)]}{d[h\nu]} = \frac{m}{(h\nu - E_g)} \quad (4)$$

Additionally, the valence and CB edge energies of a semiconductor can be determined using the following empirical formulas^{40,46}:

$$E_{VB} = X - E_e + \left(\frac{1}{2}\right) E_g \quad (5)$$

$$E_{CB} = E_{VB} - E_g \quad (6)$$

where X denotes the absolute electronegativity of a semiconductor atom. It is calculated as the geometric mean of the electronegativities of its constituent atoms. Each electronegativity is defined as the arithmetic mean of the atomic electron affinity and the first ionization energy. E_e is the free electron energy on the hydrogen scale (about 4.5 eV). E_g represents the band gap energy. E_{CB} is the conduction band potential while E_{VB} is the valence band potential. For CeO₂, X is 5.56 eV. Therefore, the estimated E_{VB} for pristine Ce_{0.8}Zr_{0.2}O₂ is 2.77 eV, with the corresponding E_{CB} at −0.65 eV.

The Urbach tail near the optical band edge signifies an exponential absorption region, typical in disordered or amorphous materials, resulting from localized states within the band gap. The absorption (α) dependence on photon energy ($h\nu$) is described by the Urbach empirical rule^{40,46}:

$$\alpha = \alpha_0 \times \exp\left(\frac{h\nu}{E_U}\right) \quad (7)$$

where α_0 is a constant, $h\nu$ is the photon energy, and E_U represents the Urbach energy, which is related to the width of the band tail caused by localized states. Taking the natural logarithm of both sides of Equation (7) yields:

$$\ln(\alpha) = \ln\alpha_0 + \left(\frac{h\nu}{E_U}\right) \quad (8)$$

To assess lattice disorder, the Urbach energy (E_U) was calculated from Equation (8), yielding $E_U = 0.93$ eV, indicative of low structural defects and a highly ordered lattice similar to published studies.^{46–48} The minimal defect states, corroborated by the oxygen vacancy data from XPS, confirm the material's pristine condition and its suitability for radiation-tolerant and high-temperature environments.

3.1.3 | Photoluminescence spectroscopy analysis

Figure 3 shows the PL spectra of pristine Ce_{0.8}Zr_{0.2}O₂ which provides essential insights into its electronic structure and defect states. The room-temperature PL spectrum excited at 250 nm highlights emission features influenced by oxygen vacancies and Ce⁴⁺ or Ce³⁺ charge states.^{33,49} The room-temperature PL spectrum excited at 250 nm shows emission peaks spanning 330–650 nm. A strong ultraviolet emission near 340 nm aligns with the bandgap energy of 3.42 eV indicates direct band-to-band radiative recombination. Blue emissions at the wavelengths of 420 nm, 445 nm, and 475 nm and green emissions at the

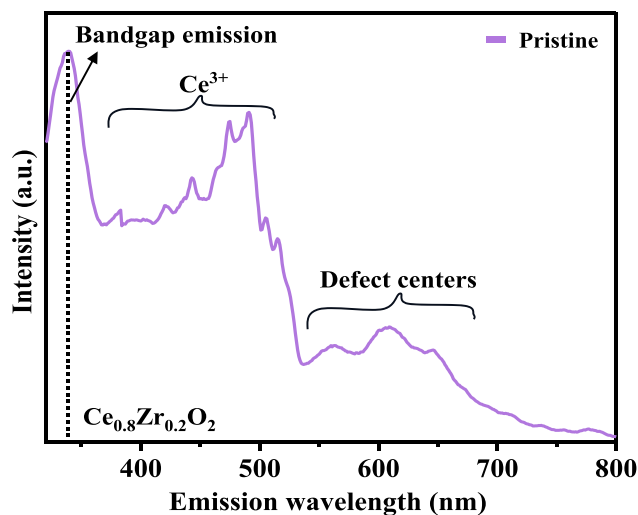


FIGURE 3 Photoluminescence spectra of pristine samples of $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$.

wavelengths of 492 nm, 505 nm, 515 nm, and 560 nm are attributed to defect levels related to oxygen vacancies and dislocations. These emissions arise from transitions between Ce 4f levels and O 2p levels.

The Ce^{3+} ions act as hole traps, while oxygen vacancies serve as electron traps, and their radiative recombination produces the observed blue and green emissions. In contrast, emission beyond 550 nm originates from defect centers that introduce electronic states within the band gap, leading to emission in various oxide matrices, as detailed in previous studies.^{41,46,49,50} Red emissions around 600–620 nm in pristine $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ are due to shallow states from extended defect clusters or interstitial oxygen atoms promoting recombination without disturbing the band structure. The low intensity of defect-related emissions compared to band-edge transitions indicates high crystallinity and minimal defects. This is confirmed by a low Urbach energy of 0.93 eV from UVDRS and the Ce^{4+} to Ce^{3+} ratio from XPS showing a low density of nonstoichiometric defects and stable optical behavior for radiation-resistant applications.

3.2 | Quantification of SHI-induced damage in $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$

3.2.1 | XPS studies of irradiated pellets

To investigate irradiation-induced structural defects, Figure 4 shows the Ce 3d XPS spectra of $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ samples irradiated with 100 MeV iodine ions at various fluences. The spectra consist of ten features due to spin-orbit splitting into Ce 3d_{5/2} (v_0 – v_4) and Ce 3d_{3/2} (u_0 – u_4). Peaks v_0 , v_2 , u_0 , and u_2 correspond to Ce^{3+} , while peaks

v_1 , v_3 , v_4 , u_1 , u_3 , and u_4 correspond to Ce^{4+} states.^{11,12,34,35} An increase in the intensity of Ce^{3+} features is evident with increasing ion fluence. Deconvolution of these spectra, as shown in Table 1, a quantitative analysis using Equation (9), reveals that the Ce^{3+} fraction increases from 27.79% in the pristine sample to 48.84% at the maximum fluence. This trend is also visualized in Figure 4F, which shows a monotonic rise in the $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio with ion fluence. These observations are consistent with previous studies on irradiated CeO_2 by Grover et al.,¹² and Iwase et al.²¹

The concentrations of Ce^{3+} and Ce^{4+} have been calculated by the v and u assignments, namely:

$$[\text{Ce}^{3+}] \text{ in } \% = \frac{A_{v_0} + A_{v_2} + A_{u_0} + A_{u_2}}{A_{v_0} + A_{v_1} + A_{v_2} + A_{v_3} + A_{v_4} + A_{u_0} + A_{u_1} + A_{u_2} + A_{u_3} + A_{u_4}}$$

$$\text{and } [\text{Ce}^{4+}] \text{ in } \% = 1 - [\text{Ce}^{3+}] \quad (9)$$

where A_i is the integrated area of peak “i.”

The progressive increase in Ce^{3+} concentration reflects partial reduction of Ce^{4+} , driven by electronic energy loss and facilitated by oxygen vacancy formation as shown schematically in Figure 7. This redox process arises from localized electron trapping under SHI irradiation. Given the extremely low oxygen diffusivity in dense ceramics at room temperature, surface reoxidation can be ruled out. As noted in the characterization section, the ~5–10 nm probing depth of XPS is sufficient to detect these near-surface chemical changes. Furthermore, the observed redox trend is in agreement with PL and UV–Vis DRS results discussed later, confirming that the Ce 3d spectra reflect broader defect evolution beyond the surface. A significant increase in the Ce^{3+} fraction in $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ on irradiation aligns with the Raman spectroscopic findings, indicating that the defect bands are linked to Ce^{3+} presence.¹² The irradiated $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ exhibits a notably higher concentration of Ce^{3+} -related defects than the pristine material.

Furthermore, Figure 5 presents the O 1s spectra and their fitted components for the pristine and irradiated $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ samples. Peak deconvolution (Table 2) identifies three distinct contributions: lattice oxygen (~529.02 eV), oxygen bonded to Ce^{3+} (~530.0 eV), and nonlattice oxygen or oxygen vacancies (~531.3 eV). Upon increasing fluence, the fraction of lattice oxygen decreases while the Ce^{3+} and oxygen vacancy components increase significantly. At the highest fluence, the lattice oxygen fraction drops to 21.23%, while Ce^{3+} -related and vacancy-associated oxygen reach 37.02% and 41.75%, respectively. These changes are also reflected in Figure 5E, which shows a clear fluence-dependent trend.

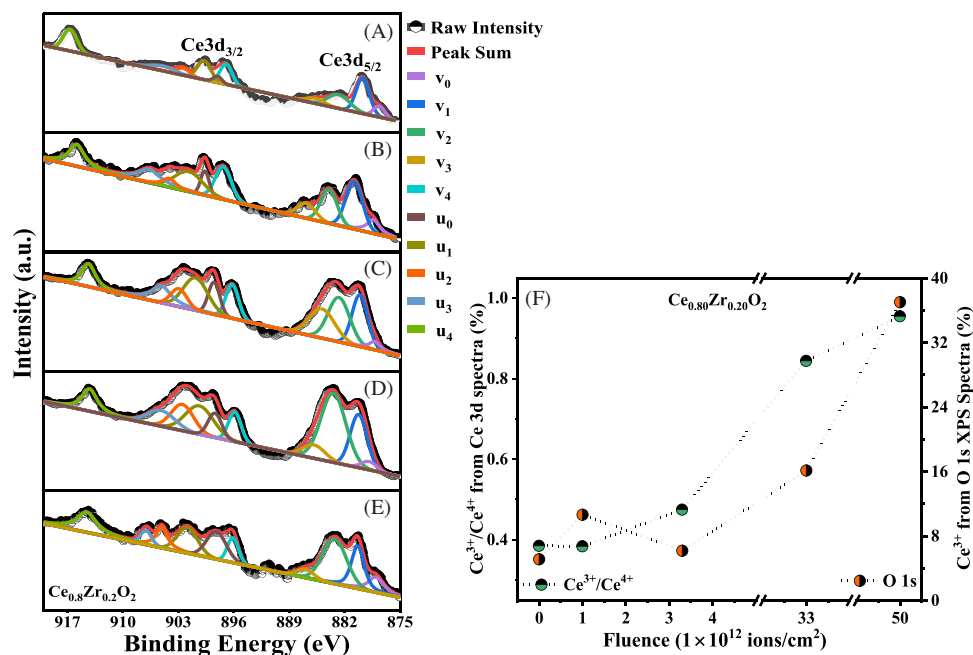


FIGURE 4 X-ray photoelectron spectroscopy (XPS) Ce 3d spectra of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, before and after being exposed to 100 MeV I^{7+} ions at fluences between from 1×10^{12} to 5×10^{13} ions/cm² (A) Pristine, irradiated at the fluence of (B) 1×10^{12} ions/cm², (C) 3.3×10^{12} ions/cm², (D) 1×10^{13} ions/cm², (E) 5×10^{13} ions/cm², and (F) $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio from Ce 3d spectra and Ce^{3+} from O 1s spectra.

TABLE 1 X-ray photoelectron spectroscopy (XPS) peaks assignment and concentration of individual Ce^{3+} and Ce^{4+} in (%) of Ce 3d before and after irradiation for polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$.

Binding energy (B.E. in eV) and peak area (P.A.) of Ce_3d												Concentration of element Ce 3d (%)	
Sample name	Ce_3d _{5/2}						Ce_3d _{3/2}						
Peaks parameter	v ₀ /Ce ³⁺	v ₁ /Ce ⁴⁺	v ₂ /Ce ³⁺	v ₃ /Ce ⁴⁺	v ₄ /Ce ⁴⁺	u ₀ /Ce ³⁺	u ₁ /Ce ⁴⁺	u ₂ /Ce ³⁺	u ₃ /Ce ⁴⁺	u ₄ /Ce ⁴⁺	Ce ³⁺	Ce ⁴⁺	
Pristine	B.E.	879.46	881.41	884.23	887.01	897.50	898.51	900.10	902.54	905.01	915.89	27.79	72.21
	P.A.	49.12	190.3	121.02	36.62	98.28	21.21	101.76	37.33	48.06	119.29		
1 × 10 ¹² ions/cm ²	B.E.	879.3	881.67	884.78	887.58	897.84	900.10	902.09	904.52	906.99	915.93	27.73	72.27
	P.A.	273.40	830.25	575.96	277.82	540.64	214.34	583.29	175.02	471.61	524.93		
3.3 × 10 ¹² ions/cm ²	B.E.	878.21	880.07	882.72	884.87	896.20	898.49	900.83	902.94	905.14	914.36	32.19	67.81
	P.A.	153.97	912.19	1081.98	841.92	484.78	413.59	891.81	321.94	550.69	471.50		
1 × 10 ¹³ ions/cm ²	B.E.	878.92	880.23	883.42	885.87	895.88	898.38	900.34	902.48	905.19	914.21	45.77	54.23
	P.A.	200.86	1431.81	2732.57	467.86	594.30	530.65	1256.49	931.38	691.46	766.41		
5 × 10 ¹³ ions/cm ²	B.E.	878.04	880.34	883.26	886.95	896.11	898.21	901.85	905.10	907.09	914.58	48.84	51.16
	P.A.	57.29	115.78	185.52	25.02	47.42	94.08	92.36	60.66	31.67	104.16		

The increase in oxygen vacancy and Ce^{3+} components with irradiation confirms a redox-driven defect formation mechanism, where Ce^{4+} is partially reduced to Ce^{3+} , and charge neutrality is maintained through oxygen vacancy formation. Despite the increase in defect concentrations, the fluorite structure remains stable, indicating a high defect tolerance facilitated by Zr incorporation. Zr promotes oxygen vacancy mobility and helps accommodate

defect stress, preserving the lattice integrity. These findings align with the Ce 3d XPS data and Raman results, which also link defect band growth to Ce^{3+} formation.^{33,34} Prior to irradiation, the presence of minor Ce^{3+} and oxygen vacancies already indicates a slightly nonstoichiometric system.

Upon irradiation, these features intensify, confirming enhanced redox activity. Grain boundaries likely

TABLE 2 X-ray photoelectron spectroscopy (XPS) binding energies (eV) and relative area (%) of the individual charge state of O1s before and after irradiation of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$.

Sample name		Binding energy (B.E. in eV) and peak area (P.A.) of O 1s			Concentration of element O 1s (%)		
		Oxygen	Ce^{3+}	Vacancies	$\text{Ce}^{4+}/\text{O}^{2-}$	Ce^{3+}	Vacancies
Pristine	B.E.	529.02	530.10		94.9	05.11	
	P.A.	1974.25	106.23				
1×10^{12} ions/cm ²	B.E.	528.92	530.34	531.82	28.49	10.64	60.87
	P.A.	1367.56	511.04	2922.01			
3.3×10^{12} ions/cm ²	B.E.	529.04	530.80	531.34	17.44	06.17	76.39
	P.A.	693.17	245.07	3036.05			
1×10^{13} ions/cm ²	B.E.	529.01	530.75	531.51	17.01	16.13	66.85
	P.A.	844.45	800.86	3318.54			
5×10^{13} ions/cm ²	B.E.	529.63	530.61	531.47	21.23	37.02	41.75
	P.A.	631.50	1101.49	1242.28			

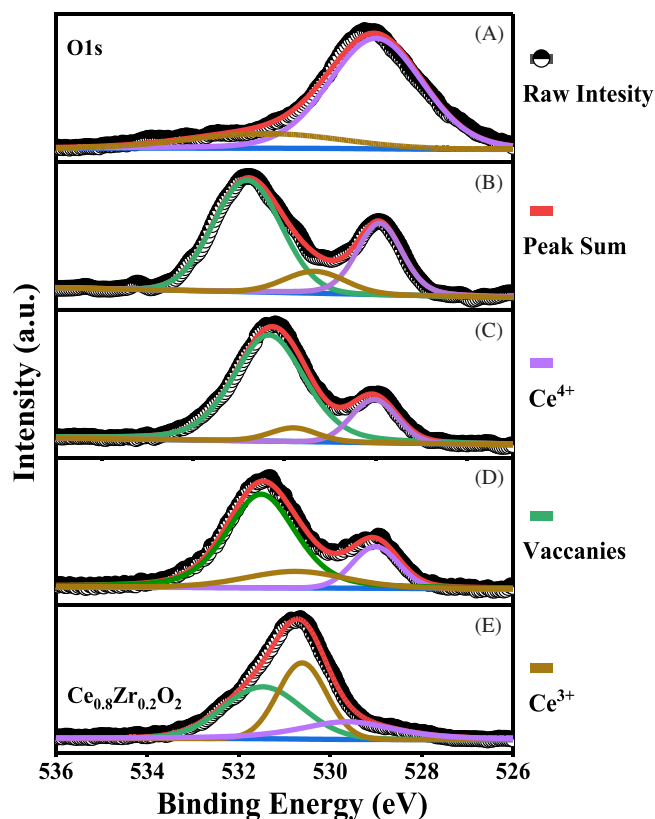


FIGURE 5 X-ray photoelectron spectroscopy (XPS) O1s spectra of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, before and after being exposed to 100 MeV I^{7+} ions at fluences between from 1×10^{12} to 5×10^{13} ions/cm² (A) pristine, irradiation at the ion fluence of (B) 1×10^{12} ions/cm², (C) 3.3×10^{12} ions/cm², (D) 1×10^{13} ions/cm², (E) 5×10^{13} ions/cm².

play a critical role in trapping radiation-induced interstitials, resulting in a vacancy-rich defect structure within grains.^{11,12,27} The overall response involves localized elec-

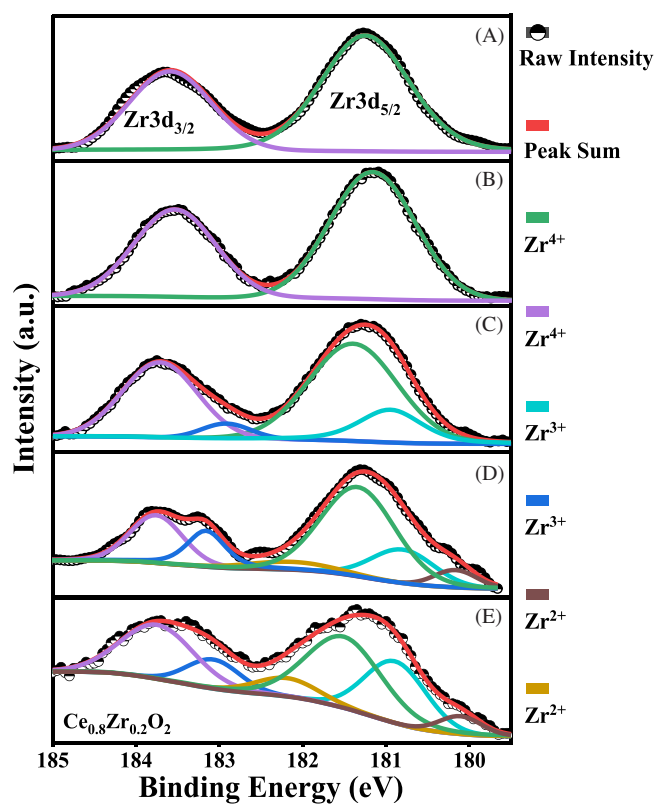
trons stabilizing the Ce^{3+} state and compensating for charge imbalance.^{51,52} During defect recovery, oxygen migration into vacancies and the reoxidation of Ce^{3+} to Ce^{4+} facilitate lattice healing.^{51,53} Oxygen vacancies thus act as defect sinks, promoting Frenkel pair recombination and suppressing irradiation-induced swelling.^{54,55} This mechanism is schematically illustrated in Figure 7 and is consistent with Ce 3d XPS trends and Raman results.

Figure 6 displays the Zr 3d XPS spectra for $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ samples irradiated with 100 MeV iodine ions at varying fluences. With increasing fluence, these peaks broaden and show asymmetry, suggesting the emergence of reduced Zr species. Deconvolution analysis (Table 3) reveals that at low fluence (1×10^{12} ions/cm²) Zr^{4+} remains stable but showing an 18% reduction to Zr^{3+} at 3.3×10^{12} ions/cm². At higher fluences (3.3×10^{13} and 5×10^{13} ions/cm²), the Zr^{3+} and Zr^{2+} fractions increase significantly, reaching 33.43% and 8.91%, respectively. Peaks at 180.3 eV and 182.9 eV correspond to the spin-orbit splitting of Zr 3d_{5/2} and Zr 3d_{3/2}, consistent with literature on irradiated zirconia systems.^{11,37–39}

SHI irradiation induces partial reduction of Zr^{4+} , Zr^{3+} , and Zr^{2+} , leading to the formation of oxygen-deficient suboxides within $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ as schematically depicted in Figure 7. This reduction is caused by electronic energy loss, which enables electron trapping at Zr^{4+} sites. The presence of multiple Zr oxidation states suggests the formation of substoichiometric phases and increasing oxygen vacancies. At higher fluence, deeper reduction to Zr^{2+} reflects stronger electronic excitation effects, as reported by Ingo et al.,³⁸ Zhang et al.,³⁷ and Liu et al.¹¹ This reduction process, driven by high-energy ion irradiation, introduces significant chemical changes, alters the oxidation state of Zr, and creates a complex defect landscape,

TABLE 3 X-ray photoelectron spectroscopy (XPS) binding energies (eV) and relative area (%) of the individual charge state of Zr 3d before and after irradiation of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$.

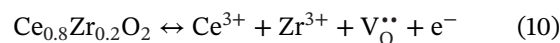
Sample name	Peaks parameter	Binding energy (B.E. in eV) and peak area (P.A.) of Zr 3d						Concentration of element Zr 3d (%)		
		Zr ⁴⁺		Zr ³⁺		Zr ²⁺		Zr ⁴⁺	Zr ³⁺	Zr ²⁺
Pristine	B.E.	Zr 3d _{5/2}	Zr 3d _{3/2}	Zr 3d _{5/2}	Zr 3d _{3/2}	Zr 3d _{5/2}	Zr 3d _{3/2}	100		
	P.A.	181.35	183.69							
1×10^{12} ions/cm ²	B.E.	181.26	183.63					100		
	P.A.	405.39	265.00							
3.3×10^{12} ions/cm ²	B.E.	181.45	183.82	180.94	183.08			81.99		
	P.A.	316.39	200.24	86.91	26.55					
1×10^{13} ions/cm ²	B.E.	181.34	183.76	180.79	183.16	180.17	182.13	64.78	25.68	09.53
	P.A.	232.78	81.57	79.27	70.03	22.31	18.16			
5×10^{13} ions/cm ²	B.E.	181.68	183.99	181.01	183.31	180.17	181.66	57.65	33.43	08.91
	P.A.	32.40	20.43	21.00	9.63	4.01	4.16			

**FIGURE 6** X-ray photoelectron spectroscopy (XPS) Zr 3d spectra of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, before and after being exposed to 100 MeV I^{7+} ions at fluences between from 1×10^{12} to 5×10^{13} ions/cm² (A) pristine, and irradiation with ion fluence of (B) 1×10^{12} ions/cm², (C) 3.3×10^{12} ions/cm², (D) 1×10^{13} ions/cm², (E) 5×10^{13} ions/cm².

thereby influencing the structural and chemical stability of $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$.

Figure 7 schematically represents the redox and defect evolution mechanism under SHI irradiation in $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$. Rather than direct formation of oxygen vacancies ($\text{V}_\text{O}^{\bullet\bullet}$) by electronic excitation, the reduction of Ce^{4+} and Zr^{4+} is primarily driven by electronic energy loss (S_e), which generates high-density electronic excitations. These excitations promote the trapping of free electrons (e^-) on Ce^{4+} and Zr^{4+} cations, leading to a $4+/3+$ equilibrium state through small polaron formation. These results are consistent with studies by Costantini et al. In ceria, they reported that Ce^{3+} formation correlates with a characteristic ~ 2.8 eV absorption band under S_e -dominated irradiation.⁵⁶ In a separate work on zirconia-based systems, they showed that Zr^{3+} formation is driven by electronic excitation and elastic collisions, leading to oxygen sublattice disorder and Frenkel pair generation.^{57,58} Therefore, in the present case, the increase in Ce^{3+} and Zr^{3+} content results from electron trapping induced by S_e , while the concurrent formation of oxygen Frenkel pairs is driven by nuclear stopping (S_n), particularly at grain boundaries. These complementary processes result in a defect-stabilized structure, enabling effective charge compensation and structural retention under irradiation.

This coupled redox response contrasts with pure CeO_2 , where only Ce reduction dominates, and with ZrO_2 , where Zr reduction is less pronounced.^{11,12,38} Spectroscopic evidence confirms that these processes are directly linked to band gap narrowing, an increase in Urbach energy, and PL quenching, consistent with polaronic charge localization and irradiation-induced lattice disorder.^{45,48,58} The equilibrium governing this process can be expressed as:



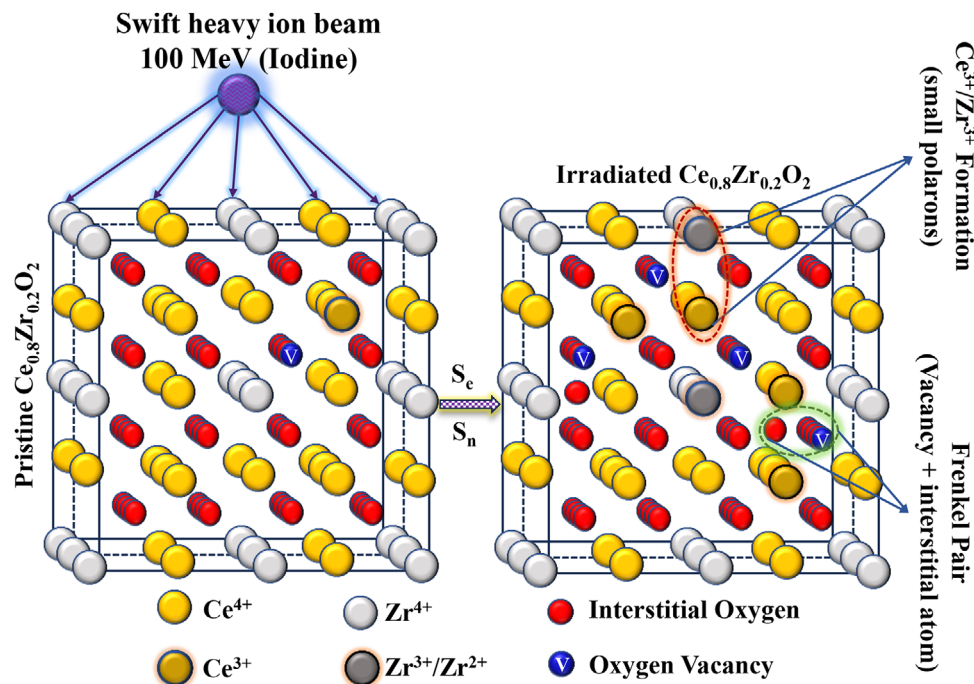


FIGURE 7 Schematic representation of oxidation state changes in ceria and zirconia with oxygen vacancies formation in $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ after irradiation with 100 MeV I^{7+} up to a fluence of 5×10^{13} ions/cm². The illustration qualitatively depicts how swift heavy ion irradiation induces Frenkel pairs via nuclear collisions and cation reduction via electron trapping from electronic excitation that influences defect stabilization. The complementary action of S_n and S_e governs defect accommodation and the retention of the fluorite structure under irradiation.

This redox–vacancy equilibrium highlights the synergy between intrinsic nonstoichiometry and irradiation-driven defects in stabilizing reduced cations and preserving the fluorite structure even at the highest fluences.^{2,12,56} Overall, the dual mechanism of electronic excitation-induced polaron formation and nuclear energy loss-driven Frenkel pair generation explains the superior radiation tolerance of $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ compared to its individual oxide counterparts.

3.2.2 | UV–Vis spectroscopy of irradiated pellets

Figure 8A presents the absorbance spectra of pristine and irradiated polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ samples. The absorption edges initially observed in the 200–450 nm range shift toward higher wavelengths as ion fluence increases. The absorption peak shifts from 386.5 to 416 nm while the edges broaden, especially at 3.3×10^{13} and 5×10^{13} ions/cm², consistent with the findings of Varshney et al.⁴³ The band gap values determined from Tauc's plot using the Schuster–Kubelka–Munk function, decrease from 3.42 eV to 2.68 eV as fluence increases from 1×10^{12} to 5×10^{13} ions/cm² (Figure 8C). To validate these results, the optical band gap was also evaluated using the derivative method

of Tauc's equation, which provided consistent values with Tauc's analysis (Figure 8D). The band gap remains near the pristine value of 3.42 eV at lower fluences of 1×10^{12} , 3.3×10^{12} , and 1×10^{13} ions/cm². However, at higher fluences of 3.3×10^{13} and 5×10^{13} ions/cm², it decreases to 2.82 eV and 2.68 eV, respectively. This behavior is consistent with previous studies of band gap narrowing in SnO_2 (3.75 eV to 3.1 eV under 75 MeV Ni^{+}),⁴ CeO_2 (3.42 eV to 3.11 eV under 175 MeV Au^{13+}),⁴³ and CdS (2.41 eV to 2.35 eV under 100 MeV Ag^{7+}).⁵⁹

The Urbach energy calculated from the linear region of the $\ln(\alpha)$ versus $h\nu$ plot using Equation (8) increases from 0.93 eV for pristine to 1.85 eV at the highest fluence (Figure 8B), alongside the increase in refractive index is also observed with increasing fluence, as summarized in Table 4.

The red shift in absorption and narrowing of the band gap result from irradiation-induced of localized defect states, particularly oxygen vacancies and Ce^{4+} to Ce^{3+} reduction. These defects introduce subbandgap states that absorb lower-energy photons, reducing the effective band gap.^{43,46,59} The Tauc method captures this trend and aligns with XPS evidence of increased Ce^{3+} and oxygen vacancy concentrations, confirming its validity for analyzing irradiation-driven electronic disorder. At lower fluences, the minor band gap reduction may be

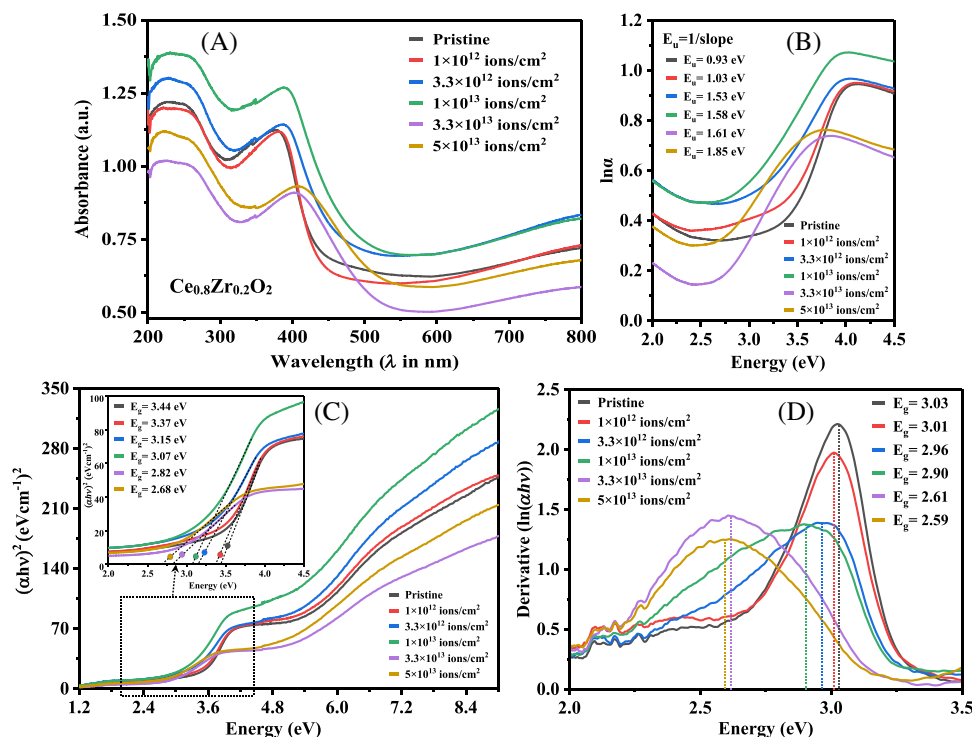


FIGURE 8 The absorption spectra of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, before and after being exposed to 100 MeV I^{7+} ions at fluences between from 1×10^{12} to 5×10^{13} ions/ cm^2 (A) UV-Vis diffuse reflectance spectroscopy (UVDRS) spectra, (B) $\ln(\alpha)$ versus $h\nu$ for Urbach energy, (C) $(\alpha h\nu)^{1/2}$ versus $h\nu$ Tauc plot, (D) Derivation method for bandgap calculation.

TABLE 4 Optical parameters for pristine and irradiated of $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ samples.

Samples	Optical bandgap E_g (eV)		Urbach energy E_U (eV)	Refractive index (n) at 500 nm
	Tauc's formula	Derivative method		
Pristine	3.42	3.03	0.93	2.24
1×10^{12} ions/ cm^2	3.36	3.01	1.03	2.25
3.3×10^{12} ions/ cm^2	3.25	2.96	1.53	2.27
1×10^{13} ions/ cm^2	3.07	2.90	1.58	2.29
3.3×10^{13} ions/ cm^2	2.82	2.61	1.61	2.31
5×10^{13} ions/ cm^2	2.68	2.59	1.85	2.34

attributed to enhanced crystallization,⁴ as supported by X-ray diffraction (XRD) results.² However, beyond 3.3×10^{13} ions/ cm^2 , accumulated defects dominate. The band gap narrowing arises from Urbach tailing and defect-induced mid-gap states. The reduction of Ce^{4+} to Ce^{3+} introduces localized 4f states about 1 eV below the CB minimum, consistent with Hubbard-U-corrected models. Such states are associated with polaron formation in ceria and with defect clustering reported in zirconia-based systems under irradiation.^{56–58,60} Together, these mechanisms explain the strong subbandgap absorption and band tailing observed at higher fluences. The rise in Urbach energy here reflects such enhanced microstruc-

tural disorder, consistent with defect generation and oxygen nonstoichiometry,^{40,46,47} while the increase in refractive index suggests densification and polarizability due to defect clustering and Ce^{3+} accumulation. These observations are supported by XPS and defect formation under irradiation.

The top of the VB and the bottom of the CB energies of a semiconductor can be determined using Equations (5) and (6). Thus, the E_{VB} for pristine and irradiated of $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ are estimated to be 2.77 eV, 2.74 eV, 2.69 eV, 2.60, 2.47 eV, and 2.40 eV, and their corresponding E_{CB} are; -0.65 eV, -0.62 eV, -0.57 eV, -0.48 eV, -0.35 eV, and -0.28 eV, respectively.

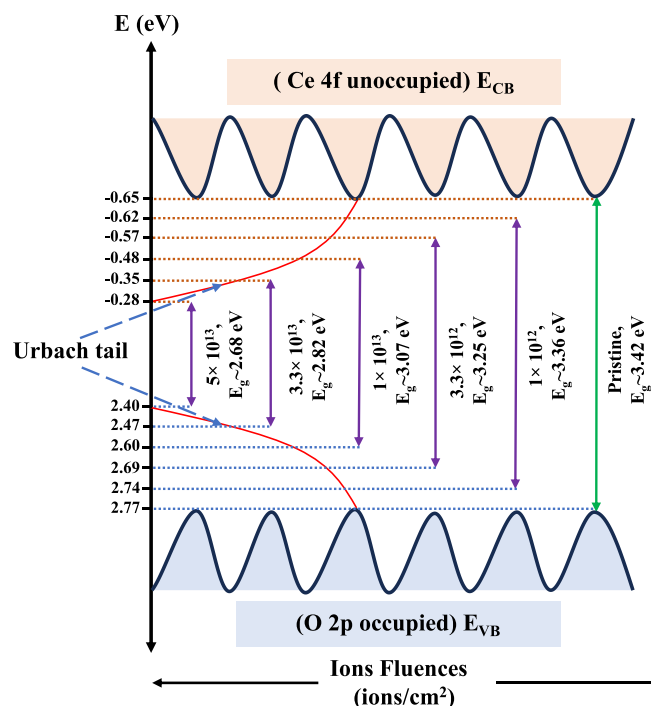


FIGURE 9 Schematic energy band diagram of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, before and after irradiation, showing band gap narrowing due to Urbach tailing, defect-induced mid-gap states, at fluences from 1×10^{12} to 5×10^{13} ions/cm².

The energy band diagram of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ reveals significant electronic modifications due to ion irradiation, depicted in Figure 9. The pristine sample's 3.42 eV direct band gap, dominated by Ce 4f and O 2p states, narrows to 2.85 eV at 5×10^{13} ions/cm², attributed to oxygen vacancies and Ce^{4+} reduction to Ce^{3+} . UVDRS and XPS data confirm these vacancies introduce defect states below the Ce 4f CB, causing red shifts in absorption spectra and increased Urbach energy from 0.93 eV to 1.85 eV arising from point defect accumulation.^{46–48} These changes enhance subgap transitions and lattice adaptability for extreme-environment applications.

3.2.3 | Photoluminescence spectroscopy of irradiated pellets

Figure 10 presents the PL spectra of pristine and irradiated polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ samples. In the pristine state, the PL spectrum is dominated by a sharp emission near 340 nm, corresponding to the 3.42 eV bandgap and indicative of direct band-to-band radiative recombination. Additionally, broad emissions in the visible region are associated with intrinsic oxygen vacancies and defect-mediated recombination pathways. Upon irradiation, new emission peaks emerge in the blue and green regions, attributed

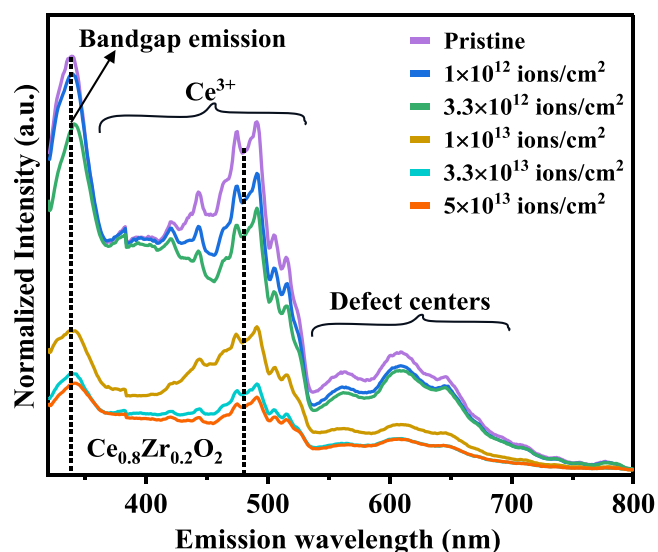


FIGURE 10 Photoluminescence spectra of polycrystalline $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$, before and after being exposed to 100 MeV I^{7+} ions at fluences ranging from 1×10^{12} to 5×10^{13} ions/cm².

to the formation of localized defect states such as oxygen vacancies and the partial reduction of Ce^{4+} to Ce^{3+} , as confirmed by XPS.^{41,49,50}

While the Ce^{3+} concentration increases with fluence, the observed PL intensity reduction is linked to the interplay of radiative and nonradiative recombination mechanisms as represented in Figure 10. Ce^{3+} ions enhance defect-related radiative transitions, but the simultaneous creation of charged oxygen vacancies introduces competing nonradiative pathways, as described in detail.^{46,50} Ce^{3+} promotes radiative transitions but charged defect clusters or oxygen vacancies introduce nonradiative pathways reducing carrier mobility and elongating recombination times. UV-Vis spectroscopy reveals bandgap narrowing and elevated Urbach energy which further enhances non-radiative defect states. At higher fluences red shifting and broadening of emission peaks reflect subgap defect states and increased electron-phonon coupling. The resulting red shift and broadening of emission peaks reflect the growing influence of defect-mediated transitions.

4 | CONCLUSION

This work demonstrates that $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ maintains its fluorite structure under SHI irradiation even at the high irradiation fluences. Spectroscopic analyses show that irradiation induces partial reduction of Ce^{4+} and Zr^{4+} ions, coupled with oxygen vacancy formation. The resulting trapped electrons localize on Ce and Zr sites, forming small polarons that stabilize the reduced cation states through local lattice distortions. This polaronic behavior

is supported by XPS and further evidenced by UV–Vis DRS, which shows a band gap reduction and significant enhancement in Urbach energy, both indicative of defect-induced localized states and electronic disorder. PL spectra show a systematic decrease in intensity with fluence, reflecting enhanced nonradiative recombination pathways via these defect–polaron complexes. The combination of oxygen Frenkel pairs (from nuclear collisions) and polaronic carriers (from electronic excitations) illustrates a dual mechanism for defect accommodation. The coupled Ce–Zr redox response, stabilized by oxygen vacancy–Frenkel pair equilibrium, enables efficient charge compensation and defect recovery, thereby preventing amorphization and highlighting $\text{Ce}_{0.8}\text{Zr}_{0.2}\text{O}_2$ as a promising material for nuclear fuels, space applications, and other extreme radiation environments.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the funding support from the Science and Engineering Research Board (SERB), Department of Science and Technology, Government of India, for the CRG/2018/003937 project. Vivek Kumar appreciates the UGC-SRF for awarding him Reference No. 1623 (CSIR-UGC NET JUNE 2019). The authors are appreciative of the stable iodine beam facility that was provided by IUAC New Delhi. The authors thank IIT Roorkee for providing an X-ray Photoelectric Spectroscopy facility. They also extend their gratitude to AIRF-JNU, New Delhi, for offering essential tools for synthesis and characterization, including high-temperature furnaces and Raman Spectroscopy.

DATA AVAILABILITY STATEMENT

The information that validates this investigation's conclusions is accessible upon reasonable request from the principal investigator.

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SUPPORTING INFORMATION

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How to cite this article: Kumar V, Sharma SK, Singh Y, Malik V, Grover V, Kulriya PK. Swift heavy ion-induced defect formation and electronic structure evolution in Zr-doped CeO₂ (Ce_{0.8}Zr_{0.2}O₂). *J Am Ceram Soc.* 2026;109:e70310. <https://doi.org/10.1111/jace.70310>